

Variety

CR Dhan 310

(IET24780: CR2829-PLN-37)

Naveen X ARC10075

Product Profile

CR Dhan 310 is the first high protein rice variety in India, released by NRRI, Cuttack for the states of Odisha, Uttar Pradesh and Madhya Pradesh. It has medium duration (120-125 days), semi-dwarf plant type (110 cm) with medium slender and good grain quality. Recommended for irrigated and favourable shallow rainfed areas. National average of grain yield is 4.5 t ha⁻¹ and it contains average 10.2% protein in polished rice.

Priority Market Segment - TLaS-R

Planting Method		Ecosystem		
PTR	DSR	Irrigated Lowland	Rainfed Lowland	Upland
Recommended	Not Recommended	Recommended	Not Recommended	Not Recommended

Positioning

Position	Irrigated Lowland	Rainfed Lowland	Upland
Replace	CR Dhan303	None	None
Companion	CR Dhan311	None	None

Key Strengths

- Contain 10.3% Protein in polished grain
- Compact and well-exserted panicle
- Medium slender grain

Considerations

- Average yield

Relative Variety Performance - Grain Yield and Agronomic Traits

Variety	Grain yield		Maturity (days)	Plant height (cm)	Lodging	1000 grain weight (g)	HRR %	Amylose content %
	(Kg/ha)	% Adv						
CR Dhan 310	4500		120-125	110	Tolerant	24	64.7	25.1
CR Dhan 303	5000		125	115	Tolerant	23.5	62.20	24.5

Relative Variety Performance - Biotic and Abiotic Stress

Variety	Neck blast	Brown spot	Sheath rot	Leaf folder	Rice tungro	Drought
CR Dhan 310	Tolerant	Tolerant	Tolerant	Tolerant	Tolerant	NA
CR Dhan 303	Tolerant	Tolerant	Tolerant	Tolerant	Tolerant	NA

Producibility: CR dhan 310: = 1800Kgs. / acre

Recommendation: Odisha, Madhya Pradesh and Uttar Pradesh

Relative Variety Performance - Biotic and Abiotic Stress



Variety

DRR Dhan 50

Samba Mahsuri sub-1*

with qDTY2.1& 3.1

Product Profile

DRR Dhan 50 was released in 2018 by IIRR, Hyderabad is similar to Samba Mashuri with a dual abiotic stress tolerance against submergence and drought. Recommended for irrigated or transplanted low land and drought prone areas. It has strong culm, erect leaves and just exerted panicle. Medium slender grain with good cooking quality. Moderately resistant to BLB and blast.

Priority Market Segment - TLAS-R

Planting Method		Ecosystem		
PTR	DSR	Irrigated Lowland	Rainfed Lowland	Upland
Recommended	Not Recommended	Not Recommended	Recommended	Not Recommended

Positioning

Position	Irrigated Lowland	Rainfed Lowland	Upland
Replace	None	Samba Mashuri	None
Companion	None	MTU 1075	None

Key Strengths

- Dual stress tolerant
- MS fine grain
- Moderately resistant to BLB and blast
- Non lodging
- Fine grain, Good cooking quality

Considerations

- Susceptible to leaf folder

Relative Variety Performance - Grain Yield and Agronomic Traits

Variety	Grain yield		Maturity (days)	Plant height (cm)	Lodging	HRR %	Amylose content %	1000 grain weight (g)
	(Kg/ha)	% Adv						
DRR Dhan 50	5500-6000	0	135-140	90-95	Tolerant	59.7	23	25.2
Samba Mahsuri	5500-6500		135	80-90	Tolerant	65.2	22.5	22.6

Relative Variety Performance - Biotic and Abiotic Stress

Variety	Leaf blast	Leafblight	Sheath blight	BPH
DRR Dhan 50	Tolerant	Tolerant	Tolerant	Tolerant
Samba Mahsuri	Resistant	Susceblible	Tolerant	Susceblible

Producibility: DRR dhan 50= 2400 Kgs. / acre

Recommendation: Released for cultivation in Andhra Pradesh, Telangana, Tarnilnadu, Karnataka, Bihar, Odisha, Chhattisgarh, Uttar Pradesh and Madhya Pradesh

*Parental line (s) from IRRI germplasm



Variety

Improved Lalat

Lalat and IRBB 60
(CRMAS 2621-7-1)



Product Profile

Improved Lalat is a medium duration (130 days), semi-dwarf variety released by NRRI, Cuttack in 2012 for cultivation under bacterial leaf blight prone areas of Odisha. It has good quality long slender grains with higher head rice recovery with an average productivity of 4.5-5.0 t/ha. This variety is recommended in irrigated ecology.

Priority Market Segment - TLAS-R

Planting Method		Ecosystem		
PTR	DSR	Irrigated Lowland	Rainfed Lowland	Upland
Recommended	Not Recommended	Recommended	Not Recommended	Not Recommended

Positioning

Position	Irrigated Lowland	Rainfed Lowland	Upland
Replace	IR-64	None	None
Companion	Bina dhan 11	None	None

Key Strengths

- Resistance against BLB
- Long slender grain
- Good yield
- Semi-dwarf variety

Considerations

- Progressive proneness to diseases and pests

Relative Variety Performance - Grain Yield and Agronomic Traits

Variety	Grain yield		Maturity (days)	Plant height (cm)	Lodging	HRR %	Amylose content %	1000 grain weight (g)
	(Kg/ha)	% Adv						
Improved Lalat	4500-5000	12	130	100	Tolerant	61	23	24
IR-64	4000-4300	8	120	100-105	Tolerant	49	25	23

Relative Variety Performance - Biotic and Abiotic Stress

Variety	Leaf blast	BLB	Sheath blight	Gall midge	Stemborer
Improved Lalat	Tolerant	Resistant	Tolerant	Resistant	Resistant
IR-64	Tolerant	Resistant	Susceptible	Tolerant	Tolerant

Producibility: Improved Lalat= 2000 Kgs. / acre

Recommendation: Released for cultivation in Odisha.



*Parental line (s) from IRRI germplasm

Variety

Cottondora Sannalu

(MTU-1010)

(Krishnaveni x IR64*)

Product Profile

A high yielding, early maturity, medium tall, long slender heavy grain mega variety grown extensively in India under Irrigated lowland conditions. Developed and released in 2000 by ANGRAU (Maruteru) by hybridization of a local variety 'Krishnaveni' with IR64, a popular rice variety from IRRI. Excellent productive tillering and sturdy stem. Susceptible to leaf and sheath blight and BPH. Tolerant to leaf blast.



Priority Market Segment - TLAS-R

Planting Method		Ecosystem		
PTR	DSR	Irrigated Lowland	Rainfed Lowland	Upland
Recommended	Not Recommended	Recommended	Not Recommended	Not Recommended

Positioning

Position	Irrigated Lowland	Rainfed Lowland	Upland
Replace	IR-64	None	None
Companion	Vijetha	None	None

Key Strengths

- Good yield
- Medium tall; Sturdy stem
- Long slender heavy grain
- Early maturity
- Tolerant to Leaf blast
- Excellent productive tillers

Considerations

- Susceptible to Leaf & Sheath blight and BPH
- Average performance in low phosphorus soil
- Prone to shattering

Relative Variety Performance - Grain Yield and Agronomic Traits

Variety	Grain yield		Maturity (days)	Plant height (cm)	Lodging	HRR %	Amylose content %	1000 grain weight (g)
	(Kg/ha)	% Adv						
Cottondora Sannalu (MTU-1010)	4000-4500	5	115-120	105-108	Tolerant	57.0	22.6	24.3
IR-64	4000-4300		115-120	100-105	Tolerant	49.1	24.5	23.1

Relative Variety Performance - Biotic and Abiotic Stress

Variety	Leaf blast	Leaf blight	Sheath blight	BLB	Drought
Cottondora Sannalu (MTU-1010)	Tolerant	Susceptible	Susceptible	Susceptible	Susceptible
IR-64	Tolerant	Resistant	Susceptible	Tolerant	Susceptible

Producibility: Cottondora Sannalu (MTU-1010) = 1600-1800Kgs. / acre

Recommendation: Released for cultivation in Andhra Pradesh, Telangana, Odisha & west Bengal. Zonal check in advance varietal trial (mid-early) in central and southern India.

*Parental line (s) from IRRI germplasm



Variety

Pushyami

(MTU-1075)

(MTU-2716 x MTU-1010)

Product Profile

A high yielding, late maturity, semi-tall, long slender grain variety grown extensively in South India. Developed by MARUTERU, and released in 2008. Highly adapted to irrigated lowlands. Good cooking quality. Tolerant to important diseases and insect pests. Moderate HRR. Good grain weight.

Priority Market Segment - TLaS-R

Planting Method		Ecosystem		
PTR	DSR	Irrigated Lowland	Rainfed Lowland	Upland
Recommended	Not Recommended	Recommended	Not Recommended	Not Recommended

Positioning

Position	Irrigated Lowland	Rainfed Lowland	Upland
Replace	Swarna	None	None
Companion	MTU-1010	None	None

Key Strengths

- Good yieldpotential
- Non-lodging
- Long slender grain
- Semi-tall
- Tolerant to major diseases and insect pests
- Good cooking quality

Considerations

- Susceptible to Sheath blight

Relative Variety Performance - Grain Yield and Agronomic Traits

Variety	Grain yield		Maturity (days)	Plant height (cm)	Lodging	HRR %	Amylose content %	1000 grain weight (g)
	(Kg/ha)	% Adv						
Pushyami	5500-6000	=	135-140	105-108	Tolerant	54	24	21
Swarna	5500-6000		140-145	85-90	Susceptible	69.0	26.1	16

Relative Variety Performance - Biotic and Abiotic Stress

Variety	Leaf blast	Leaf blight	Sheath blight	BPH	Gall midge	Submergence
Pushyami	Tolerant	Tolerant	Susceptible	Tolerant	Tolerant	Tolerant
Swarna	Susceptible	Resistant	Susceptible	Susceptible	-	-

Producibility: Variety Pushyami=2200-2400Kgs. / acre

Recommendation: Released for cultivation in Andhra Pradesh, Telangana, Odisha.

Zonal VII(S) check for Initial varietal trials conducted by IIRR, India.



Variety

Pooja

(Vijaya x T141)

Product Profile

A high yielding, late maturity, semi-tall, long slender grain variety grown extensively in India. Developed by NRRI, Cuttack and released in 2008. Highly adapted to irrigated lowlands. Good cooking quality. Tolerant to important diseases and insect pests. Moderate HRR. Good grain weight.

Priority Market Segment - TLAM-R

Planting Method		Ecosystem		
PTR	DSR	Irrigated Lowland	Rainfed Lowland	Upland
Recommended	Not Recommended	Not Recommended	Recommended	Not Recommended

Positioning

Position	Irrigated Lowland	Rainfed Lowland	Upland
Replace	None	Swarna	None
Companion	None	Swarna Sub-1; Khandagiri	None

Key Strengths

- High yield
- Late maturity; Profuse tillering
- Strong straw strength
- Tolerant to lodging and shattering
- Fine grain with good cooking & eating quality

Considerations

- Susceptible to Brown spot, Bakanae
- Moderate HRR
- Low grain weight

Relative Variety Performance - Grain Yield and Agronomic Traits

Variety	Grain yield		Maturity (days)	Plant height (cm)	Lodging	HRR %	Amylose content %	1000 grain weight (g)
	(Kg/ha)	% Adv						
Pooja	5000-6000	-	150-155	90-95	Tolerant	53.5	22.0	13.5
Swarna	5500-6000	-	140-145	85-90	Susceptible	69.0	26.1	15.9

Relative Variety Performance - Biotic and Abiotic Stress

Variety	Leaf blast	Leaf blight	Sheath blight	BPH	Gall midge	Submergence
Pooja	Tolerant	Tolerant	Susceptible	Susceptible	Resistant	Tolerant
Swarna	Susceptible	Resistant	Susceptible	Susceptible	Susceptible	Susceptible

Producibility: Pooja=2000-2400 Kgs. / acre

Recommendation: Released for cultivation in Odisha, Andhra Pradesh, Madhya Pradesh, West Bengal. Zonal-III (E) check for varietal trials conducted by IIRR, India



Variety

Sahbhagi Dhan

(IR 55419-4*2* x WAY RAREM*)

Product Profile

Early maturity, semi-dwarf, long-bold grain, widely adaptable variety mainly grown as direct seeded under rainfed and upland conditions. Developed by CRRI, Hajaribag & IRRI and released in 2008 for Jharkhand and Odhisa. Tolerant to drought and highly adaptable to water scarce and upland environments. Excellent grain and eating quality with low glycemic index.

Priority Market Segment - TLAM-R

Planting Method		Ecosystem		
PTR	DSR	Irrigated Lowland	Rainfed Lowland	Upland
Recommended	Recommended	Not Recommended	Recommended	Recommended

Positioning

Position	Irrigated Lowland	Rainfed Lowland	Upland
Replace	None	IR-64	IR-64
Companion	None	Samba Masuri, DRR Dhan 42	DRR Dhan 42

Key Strengths

- Drought tolerant
- Semi-dwarf; Profuse tillering; High straw yield
- Long bold grain
- Early maturity; Non-lodging
- High HRR; Excellent eating quality
- Tolerant to Leaf blast, Sheath blight & Leaf folder

Considerations

- Susceptible to low temperature
- Low yield under severe drought
- Susceptible to Leaf blight

Relative Variety Performance - Grain Yield and Agronomic Traits

Variety	Grain yield		Maturity (days)	Plant height (cm)	Lodging	HRR %	Amylose content %	1000 grain weight (g)
	(Kg/ha)	% Adv						
Sahbhagi Dhan	3800-4500	5	100-110	85-90	Tolerant	64.7	24.7	22.3
IR-64	4000-4300	-	115-120	100-105	Tolerant	49.1	24.5	23.1

Relative Variety Performance - Biotic and Abiotic Stress

Variety	Leaf blast	Leaf blight	Sheath blight	BPH	Stem borer	Leaf folder	Drought
Sahbhagi Dhan	Tolerant	Susceptible	Tolerant	Susceptible	Tolerant	Tolerant	Tolerant
IR-64	Tolerant	Resistant	Susceptible	Tolerant	Susceptible	Susceptible	Susceptible

Producibility: Sahbhagi Dhan = 1500-1800Kgs. / acre

Recommendation: Released for cultivation in Odhisa, Madhya Pradesh, Andhra Pradesh, Jharkhand, Chhattisgarh, Bihar, West Bengal, Assam, Uttar Pradesh.

National check in varietal trial (Early direct seeded) conducted by IIRR, India.



*Parental line (s) from IRRI germplasm

Variety

Swarna Sub-1

(IR 82809:227* x SWARNA)

Product Profile

Late maturity, medium height, short bold grain, high yielding widely adaptable variety grown in India under rainfed lowland conditions. Developed by NRRI and released in 2009 for cultivation in the states of Eastern India. Tolerant to complete submergence upto 15-17 days. Highly adaptable to flood prone shallow lowlands. Excellent tillering. Low nitrogen responsive. Tolerant to low soil phosphorus levels. Excellent grain and eating quality with low glycemic index.

Priority Market Segment - TLaS-R

Planting Method		Ecosystem		
PTR	DSR	Irrigated Lowland	Rainfed Lowland	Upland
Recommended	Not Recommended	Not Recommended	Recommended	Not Recommended

Positioning

Position	Irrigated Lowland	Rainfed Lowland	Upland
Replace	None	Swarna	None
Companion	None	Pooja, CR 1009 Sub-1	None

Key Strengths

- Goodyield
- Medium height; Profuse tillering
- Submergence tolerance
- Latematurity; Short bold grain
- High HRR; Excellent eating quality
- Resistant to Leaf blight

Considerations

- Susceptible to Leaf blast, Sheath blight and False smut
- Susceptible to BPH, Stem borer and Gall midge
- Loss of golden hull color as is in Swarna

Relative Variety Performance - Grain Yield and Agronomic Traits

Variety	Grain yield		Maturity (days)	Plant height (cm)	Lodging	HRR %	Amylose content %	1000 grain weight (g)
	(Kg/ha)	% Adv						
Swarna Sub-1	5000-5500	(8)	140-145	95-105	Tolerant	66.5	24.7	20.3
Swarna	5500-6000		140-145	85-90	Susceptible	69.0	26.1	15.9

Relative Variety Performance - Biotic and Abiotic Stress

Variety	Leaf blast	Leaf blight	Sheath blight	BPH	Gall midge	Submergence
Swarna Sub-1	Susceptible	Resistant	Susceptible	Susceptible	Susceptible	Tolerant
Swarna	Susceptible	Resistant	Susceptible	Susceptible	Susceptible	Susceptible

Producibility: Swarna Sub-1 = 2000-2200 Kgs. / acre

Recommendation: Released for cultivation in Odisha, Madhya Pradesh, Andhra Pradesh, West Bengal, Assam.

National check in advanced varietal trial (late) conducted by IIRR, India.



*Parental line (s) from IRRI germplasm